

This approach has a few weaknesses. First, it does not provide an easy way to make assignments when new houses are built; every newly built house has to go through the same laborious process of comparison with every existing post office. Second, doing calculations at the individual house level does not allow us to learn about a region as a whole. For example, maybe an entire neighborhood lies within the shadow of one post office but lies many miles away from all other post offices. It would be best to conclude in one step that the whole neighborhood should be served by the same close post office. Unfortunately, our method requires us to repeat the calculation for every house in the neighborhood, only to get the same result each time.

By calculating distances for each house individually, we're repeating work that we wouldn't have to do if we could somehow make generalizations about entire neighborhoods or regions. And that work will add up. In megacities of tens of millions of inhabitants, with many post offices and quick construction rates like we see today around the world, this approach would be unnecessarily slow and computing-resource-heavy.

A more elegant approach would be to consider the map as a whole and separate it into distinct regions, each of which represents one post office's assigned service area. By drawing just two straight lines, we can accomplish that with our hypothetical town ([Figure 7-3](#)).

The regions we have drawn indicate areas of closest proximity, meaning that for every single house, point, and pixel, the closest post office is the one that shares its region. Now that the entire map is subdivided, we can easily assign any new construction to its closest post office simply by checking which region it's in.

A diagram that subdivides a map into regions of closest proximity, as ours does, is called a *Voronoi diagram*. Voronoi diagrams have a long history going all the way back to René Descartes. They were used to analyze water pump placement in London to provide evidence for how cholera was spread, and they're still used in physics and materials science to represent crystal structures. This chapter will introduce an algorithm for generating a Voronoi diagram for any set of points, thereby solving the postmaster problem.